

HR ANALYTICS

KPI Values & Analysis Report

Submitted by: Dodda Subhash Reddy

Datamites Data Analytics Internship Program

B.Tech AI & ML | Krishna Chaitanya Institute of Technology & Sciences

Tool: Power BI | Dataset: HR_analysis.csv | June 2026

1. Key Performance Indicators (KPIs)

Three KPI cards were created in Power BI to provide an at-a-glance workforce summary:

KPI	Value	Supporting Data	DAX Measure
Total Employees	12,593	Full workforce headcount	COUNTROWS('HR_analysis')
Active Employees	10,367 (82.3%)	Currently employed staff	COUNTROWS(FILTER(..., termdate=BLANK()))
Terminated Employees	2,226 (17.7%)	Above 10-13% industry avg	COUNTROWS(FILTER(..., termdate<>BLANK()))
HQ Workers %	75.18%	9,468 at Headquarters	DIVIDE(HQ Count, Total)*100
Remote Workers %	24.81%	3,124 work remotely	DIVIDE(Remote Count, Total)*100
Average Age	38.4 years	Mid-career workforce	AVERAGE('HR_analysis'[Age])

2. Line Chart — Hiring Rate by Year

A line chart was built using the hire_date column, extracted to year. The DAX column used:

```
Hire Year = YEAR('HR_analysis'[hire_date])
```

Year	Hires	Year	Hires	Year	Hires
2000	118	2007	611	2014	608
2001	635	2008	628	2015	606
2002	616	2009	636	2016	648
2003	663	2010	610	2017	625
2004	660	2011	624	2018	629
2005	630	2012	611	2019	602
2006	641	2013	599	2020	592

Key Observations:

- 2000: Only 118 hires — company's first operational year (anomaly)
- 2001-2020: Stable 599-663 hires per year — no major growth phase
- 2020: Dip to 592 — reflects COVID-19 pandemic hiring freeze
- Peak years: 2003 (663), 2004 (660), 2016 (648)

3. Donut Chart — Gender Distribution

Gender	Count	Percentage	Insight
Male	6,444	51.17%	Slight majority
Female	5,808	46.12%	Near parity — positive sign
Non-Conforming	340	2.70%	Inclusive policy in place

Note: Gender column was added in Excel by deriving from first names before importing to Power BI.

4. Bar Charts

4.1 Employees by Department

Department	Employees	% of Total	Observation
Engineering	3,812	30.3%	Largest — tech-heavy focus
Accounting	1,860	14.8%	Second largest
Human Resources	1,063	8.4%	Adequate
Sales	1,019	8.1%	Moderate
Services	954	7.6%	Moderate
Training	944	7.5%	Well staffed
Business Development	918	7.3%	Moderate
Research & Development	618	4.9%	Small for tech company
Support	562	4.5%	Moderate
Product Management	371	2.9%	Small
Marketing	270	2.1%	❑ Critically understaffed
Legal	168	1.3%	Small
Auditing	33	0.3%	Very small

4.2 Employees by Race

Race	Count	% of Total
White	3,558	28.3%
Asian	2,066	16.4%
Two or More Races	2,049	16.3%

Black or African American	2,008	16.0%
Hispanic or Latino	1,484	11.8%
American Indian or Alaska Native	745	5.9%
Native Hawaiian or Other Pacific Islander	682	5.4%

4.3 Employees by State

State	Employees	% of Total	Note
Ohio	10,221	81.2%	☐ Dominant — major concentration risk
Pennsylvania	615	4.9%	Satellite location
Illinois	496	3.9%	Satellite location
Michigan	404	3.2%	Satellite location
Indiana	392	3.1%	Satellite location
Kentucky	251	2.0%	Satellite location
Wisconsin	213	1.7%	Satellite location

5. Key Insights

5.1 High Attrition Rate

At 17.7% (2,226 of 12,593), attrition exceeds the 10-13% industry benchmark by ~5 points. This indicates potential issues with compensation, management effectiveness, or career growth opportunities. Immediate root cause analysis is required.

5.2 Critical Ohio Concentration

81.2% of the total workforce (10,221 employees) is concentrated in Ohio. This single-state dependency creates major operational risk — any regional disruption could affect over 4 out of 5 employees.

5.3 Stable but Stagnant Hiring

Hiring has remained flat at 599-663 per year from 2001-2020. The company has maintained steady hiring but shows no growth phase. The 2020 dip (592) reflects COVID-19 impact.

5.4 Engineering-Heavy, Marketing-Thin

Engineering accounts for 30.3% (3,812) of all employees while Marketing has only 270 (2.1%). The 14:1 Engineering-to-Marketing ratio suggests the company builds well but may struggle to commercialize and grow its customer base.

5.5 Near Gender Parity

The 51.2% / 46.1% Male-to-Female split is a strong positive indicator. However, equity should be verified at department and seniority levels where imbalances are more common in tech-heavy organizations.

5.6 Racial Diversity Present

All 7 racial groups are represented. White employees form the largest group (28.3%) but no group is overwhelmingly dominant. Native Hawaiian/PI (5.4%) and American Indian/Alaska Native (5.9%) are smallest — targeted DEI programs are recommended.

5.7 Mid-Career Workforce — Succession Risk

The average age of 38.4 years suggests a mid-career workforce. Within 10-15 years, a significant portion of senior employees will approach retirement. Without a succession plan, critical institutional knowledge may be lost.

6. Recommendations to the Company

#	Area	Recommendation	Priority
1	Attrition	Conduct exit interviews; implement retention bonuses in high-attrition departments; target <13% attrition rate	HIGH
2	Remote Work	Increase remote/hybrid work from 25% to 40%+ to attract talent beyond Ohio's geographic boundaries	HIGH
3	Marketing	Hire 300-500 Marketing staff to close the 14:1 Engineering-to-Marketing gap and scale commercial operations	HIGH
4	Geographic Risk	Open satellite offices or enable remote hiring in underrepresented states to reduce 81% Ohio dependency	MEDIUM
5	DEI Programs	Launch targeted hiring programs for Native Hawaiian and American Indian groups; set measurable diversity KPIs	MEDIUM
6	Succession Planning	Identify high-potential junior employees; implement mentorship and knowledge transfer programs now	MEDIUM
7	Gender Equity	Conduct dept-level gender analysis; set targets for female representation in Engineering leadership roles	LOW

7. Conclusion

This HR Analytics project successfully analyzed 12,593 employee records from the HR_analysis dataset using Power BI. The dashboard includes 3 KPI cards, a line chart for hiring trends, a donut chart for gender distribution, and 3 bar charts for departments, race, and state.

The most critical findings are: (1) 17.7% attrition rate exceeding industry benchmarks, (2) 81.2% workforce concentration in Ohio creating operational risk, and (3) Marketing being critically understaffed at only 270 employees vs 3,812 in Engineering.

The Power BI dashboard built for this project provides interactive, real-time visibility into all these metrics, enabling HR managers to make data-driven decisions without requiring technical expertise.